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From Fidelity to Semantics: A Taxonomy of XAI Evaluation Metrics and Paired Empirical Comparison of LIME versus SHAP

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Abstract

Evaluation of post-hoc explanations in machine learning remains fragmented across incompatible metric families, yielding method comparisons that are sensitive to which endpoints are chosen. We address this through two complementary contributions. First, drawing on a 44-paper structured scoping corpus assembled through a documented five-database search and transparent screening record, we build a four-axis taxonomy—by evaluation target, evidence source, quality property, and task context—that maps the measurement landscape and exposes three recurring gaps: proxy metrics dominate despite not capturing semantics; human-grounded constructs remain underspecified; and semantic evaluation is growing faster than its empirical validation base. Second, we fill the proxy-layer gap with a paired empirical benchmark comparing LIME and SHAP across 75 matched cells spanning five model families, five seeds, and three sampling sizes on the Adult tabular benchmark. Within this protocol, results are consistent and large in magnitude: SHAP dominates all measured quality endpoints (stability d_z=3.00, fidelity d_z=4.82, faithfulness gap d_z=2.63), while LIME is faster for non-tree model families (median 53.3 ms vs.694.6 ms for SHAP); for tree-based models, SHAP's extttTreeExplainer reverses this ordering. A tabular cross-dataset extension on two additional datasets provides partial support for the fidelity direction, but does not establish cross-modal generality. Together, the taxonomy and the benchmark motivate a model-architecture-conditioned deployment pattern for tabular classification: SHAP is preferred at both fidelity and latency for tree-based models, while LIME retains a latency advantage for black-box architectures lacking a model-aware explainer. All experimental artifacts are publicly available.

Keywords: explainable AI, evaluation metrics, taxonomy, LIME, SHAP

1 Introduction

Selecting an explanation method in production can be posed as a constrained decision problem: given a fixed predictor f and instance x, choose $E \in \{\text{LIME}, \text{SHAP}\}$ to maximize explanation quality while satisfying operational constraints. Concretely, we consider

$$E^\star = \arg \max_{E\in\{\text{LIME,SHAP}\}} U(S_E, F_E, \Delta_{k,E}, P_E) \quad \text{s.t.} \quad C_E \leq \tau,$$

where S is stability, F fidelity, Δ_k faithfulness gap, P sparsity, C cost, and τ a latency budget. Yet this decision problem cannot be resolved in the abstract: it requires an organizing framework that identifies which quality properties matter, what evidence supports their measurement, and where structural proxy metrics end and semantic or human-centered evaluation begins.

Progress in explainable AI (XAI) depends on evaluating whether explanations are useful, faithful, stable, and meaningful. Yet the literature does not converge on a single definition of explanation quality. Surveys consistently note that XAI evaluation is fragmented across proxy metrics, qualitative user studies, and domain-specific desiderata (Adadi and Berrada, 2018; Arrieta et al., 2020; Kadir et al., 2023; Canha et al., 2025). As a result, comparisons between explainers often depend as much on the chosen metric family as on the methods themselves.

This fragmentation is especially visible in model-agnostic post-hoc explainability. Faithfulness-style metrics remain common because they are relatively easy to compute, but they do not fully capture whether an explanation is understandable, semantically aligned with domain knowledge, or decision-useful in practice (Ali et al., 2023; Longo et al., 2024). Multi-metric work shows that stability, sparsity, and cost are also part of explanation quality, and recent analyses confirm that distinct metrics reveal complementary signals (Pawlicki et al., 2024; Zheng et al., 2025).

LIME and SHAP are the most widely deployed model-agnostic post-hoc explainers for tabular data, yet controlled multi-metric comparisons under matched conditions remain scarce. Retzlaff et al. (2024) survey post-hoc versus ante-hoc design guidelines and review method properties, but do not provide matched empirical benchmarks across model families, metrics, and sampling regimes; this paper fills that gap. LIME explains locally by fitting a sparse surrogate around x:

$$\xi(x) = \arg\min_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g),$$

favoring local fidelity and explicit sparsity controls (Ribeiro et al., 2016). SHAP explains predictions through additive feature attributions $f(x) = \phi_0 + \sum_i \phi_i$, where ϕ_i are Shapley-based contributions with game-theoretic guarantees (Lundberg and Lee, 2017). Their theoretical differences are well known, but do not by themselves resolve deployment choices across model families, sampling regimes, and runtime constraints (Retzlaff et al., 2024; Ali et al., 2023).

This paper makes five contributions:

1. **Four-axis taxonomy**: we organize model-agnostic XAI evaluation along evaluation target, evidence source, quality property, and task context, making explicit why metric disagreements arise.

2. **Gap analysis**: from a documented 44-paper scoping corpus assembled through a five-database search and explicit inclusion criteria, we identify three recurring gaps—proxy dominance, underspecified human constructs, and semantic evaluation outpacing its validation base—and provide, to our knowledge, the first multi-rater LLM-judge reliability measurement in XAI (ICC<0.75 across all seven semantic dimensions). We do not claim human-subject or task-outcome validation in this paper.

3. **Formalized deployment decision**: we pose LIME-vs-SHAP selection as a constrained multi-objective problem over quality, parsimony, and cost.

4. **Paired inferential evidence**: we quantify method differences on 75 matched Adult Income cells with Holm-corrected significance testing and effect sizes for five endpoints, and partially probe tabular fidelity transfer across two additional datasets via cross-dataset validation (EXP3).

5. **Layered deployment architecture**: we derive deployment recommendations from both the taxonomy and the empirical results, mapping system constraints to explainer choice.

2 Background and Definitions

2.1 Key Distinctions

Four terminological distinctions are frequently conflated in XAI writing; keeping them separate is what makes the taxonomy and benchmark precise (Table 1). This survey and benchmark focus exclusively on post-hoc model-agnostic explainability at the local level, which is the dominant paradigm for tabular ML deployments; metric validity conditions for intrinsic models and global summaries differ substantially and are out of scope here.

2.2 LIME: Local Interpretable Model-Agnostic Explanations

LIME approximates a black-box model f locally around an instance x by training an interpretable surrogate g on perturbed samples weighted by kernel π_x:

$$\xi(x) = \arg\min_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g),$$

where $\Omega(g)$ penalizes complexity (Ribeiro et al., 2016). LIME behavior is strongly shaped by neighborhood construction hyperparameters (kernel width, perturbation sample count, feature budget). These controls yield compact explanations and low latency, but they introduce stochastic variation because local neighborhoods are sampled rather than exhaustively enumerated.

2.3 SHAP: Shapley Additive Explanations

SHAP attributes prediction output to features based on marginal contribution across all possible coalitions, satisfying efficiency, symmetry, and dummy axioms (Lundberg and Lee, 2017). KernelSHAP estimates Shapley values with a weighted regression scheme over coalitions for model-agnostic use; tree-specific implementations exploit model structure for speed. Relative to local-surrogate methods, SHAP is typically denser and computationally heavier in non-tree settings because many coalition evaluations are required.

2.4 Benchmark Design Implications

For a fair LIME-versus-SHAP comparison, confounders unrelated to explainer design must be controlled. This motivates the protocol choices in Section 5: identical transformed feature space, shared model artifacts, fixed class target, and matched cells by $(\text{model}, \text{seed}, N)$. Under these controls, expected methodological trade-offs become testable: LIME should favor sparsity and speed, while SHAP should favor stability and faithfulness. This framing

Table 1: Terminological distinctions underlying the taxonomy and benchmark scope.

Distinction	Definition	Consequence if conflated
Intrinsic vs. post-hoc	Intrinsic models are interpretable by construction; post-hoc methods explain black-box behavior after training (Rudin, 2019; Retzlaff et al., 2024).	Intrinsic-model metric validity conditions differ substantially from post-hoc ones and do not transfer.
Explanation quality vs. model quality	A model may be accurate while producing unstable or semantically weak explanations, or vice versa.	Evaluation must specify whether the object under study is the explanation artifact, explainer method, predictive model, or user-facing process; conflating levels produces inconsistent benchmark conclusions.
Local vs. global	Local attributions/counterfactuals are judged via faithfulness, sparsity, or actionability; global summaries via coverage, consistency, or conceptual coherence.	Many reported metric disagreements arise from mixing local and global levels without explicit task framing.
Semantic vs. structural	Structural proxies (e.g., perturbation faithfulness) track model sensitivity under an intervention; semantic judgments assess whether an explanation "makes sense" under domain, causal, or linguistic criteria.	The two are not interchangeable; substituting one for the other conceals gaps in evaluation coverage.

aligns with calls for multi-metric, functionally grounded XAI evaluation (Pawlicki et al., 2024; Canha et al., 2025).

3 Taxonomy of XAI Evaluation Metrics

The central contribution of this section is that XAI evaluation can be clarified by crossing four axes: evaluation target, evidence source, quality property, and task context. Metrics that appear to answer the same question often do not, because they attach to different objects, use different evidence, or imply different validity conditions.

3.1 Taxonomy by Evaluation Target

Many metric disagreements start with a hidden mismatch about what is being evaluated:

- **Explanation artifact metrics** evaluate the produced explanation itself—attribution sparsity, local fidelity, or semantic clarity.

- **Explainer method metrics** evaluate the behavior of the explainer across cases—stability, reproducibility, or computational cost.
- **Model-behavior metrics** evaluate whether explanations track model behavior under perturbations or alternative decision regions.
- **User/task outcome metrics** evaluate whether explanations improve decision performance, trust calibration, or actionability.

This distinction matters because one metric family cannot automatically stand in for another. A sparse explanation can still be semantically misleading; a faithful attribution can still be unusable in a decision-support workflow.

3.2 Taxonomy by Evidence Source

The second axis concerns the evidence used to justify the evaluation:

- **Proxy or perturbation metrics** infer quality from numerical sensitivity behavior under intervention.
- **Benchmark-grounded metrics** compare explanations to synthetic or domain-defined ground truth.
- **Human expert judgments** evaluate plausibility, correctness, or domain alignment.
- **End-user studies** assess comprehension, trust calibration, decision quality, or cognitive load.
- **LLM judges** provide scalable semantic scoring via rubric-guided language-model evaluation.

Proxy metrics are scalable and reproducible but may under-capture meaning. Human judgments are richer but expensive and variable. LLM judges are scalable like proxies but semantically expressive like human reviews, which is precisely why their validity must be established rather than assumed (Gu et al., 2024; Wu et al., 2025; Zhou et al., 2025). Evidence Source thus characterizes *how* measurement evidence is generated—the modality of observation—while Quality Property (below) characterizes *what* construct is claimed; the same quality property can in principle be assessed via multiple evidence sources, and the same source type can serve multiple properties.

3.3 Taxonomy by Quality Property

The literature returns repeatedly to a familiar set of quality properties, though not always with consistent definitions:

- **Fidelity / faithfulness:** whether the explanation tracks how the model actually behaves under feature interventions or masking (Ribeiro et al., 2016; Lundberg and Lee, 2017; Zheng et al., 2025).
- **Stability / robustness:** whether similar inputs yield similar explanations under perturbation or reruns (Pawlicki et al., 2024; Canha et al., 2025).

Table 2: Working construct dictionary: operational meanings and strongest direct evidence for each quality property.

Construct	**Operational meaning**	**Strongest evidence**
Fidelity / faithfulness	Whether the explanation tracks model behavior under a specified intervention or masking scheme.	Proxy tests and benchmark-grounded checks
Stability / robustness	Whether explanations remain consistent across reruns, perturbations, or nearby inputs.	Proxy perturbation studies and repeated-run analyses
Sparsity / parsimony	Whether explanations keep feature load low enough to remain cognitively manageable.	Structural summaries of explanation size or weight concentration
Plausibility	Whether explanations align with expert expectations or domain logic.	Expert-human review with explicit rubrics
Semantic alignment	Whether explanations capture meaningful conceptual or causal structure beyond structural proxy success.	Expert review or validated semantic evaluators
User usefulness	Whether explanations improve comprehension, decision quality, or trust calibration in an actual task.	End-user studies with task outcomes

- **Sparsity / parsimony:** whether the explanation is concise enough to be interpretable without unnecessary feature load.

- **Plausibility:** whether the explanation aligns with domain expectations or expert reasoning.

- **Counterfactual or recourse quality:** whether explanation-linked interventions are feasible, minimal, diverse, or actionable (Mothilal et al., 2020).

- **Semantic alignment:** whether an explanation captures meaningful causal or conceptual structure, even when proxy metrics are agnostic to that structure.

These properties are only partly commensurable. A method can score strongly on fidelity while remaining weak on sparsity or semantic adequacy. Multi-metric reporting is therefore not optional; it is the minimum needed to make XAI claims interpretable across use cases.

Table 3: Core metric families in model-agnostic XAI evaluation: what each captures well and what it leaves unresolved.

Metric family	**Captures well**	**Leaves unresolved**
Fidelity / faithfulness	Whether explanations track model behavior under interventions or masking.	Whether the explanation is meaningful, plausible, or useful outside the proxy setup.
Stability / robustness	Whether explanations remain consistent across perturbations or repeated runs.	Whether stable explanations are actually correct or semantically adequate.
Sparsity / parsimony	Whether explanations are concise and low in feature load.	Whether concision hides necessary nuance or context.
Benchmark-grounded correctness	Whether explanations recover synthetic or domain-defined ground truth.	Whether the ground truth is realistic, complete, or transferable.
Human expert review	Whether explanations align with domain expectations and expert reasoning.	Whether those judgments are scalable, standardized, and reproducible.
User-centered evaluation	Whether explanations improve understanding, decisions, or trust calibration in practice.	Whether findings generalize across tasks, populations, and study designs.
LLM-based semantic evaluation	Whether rubric-based semantic judgments can be produced quickly and at scale.	Whether those judgments agree with humans and avoid proxy-judge bias.

3.4 Taxonomy by Task Context

Metric validity also depends on context. Tabular benchmarks often assume feature independence, perturbation realism, and tractable local neighborhoods. These assumptions can fail in image, text, graph, and time-series settings, where perturbations may create out-of-distribution examples or semantically invalid transformations (Agarwal et al., 2023; Proszewska et al., 2025). Metric transport across modalities cannot be assumed.

Table 3 illustrates the practical consequence of the taxonomy: each metric family gains strength by focusing on one kind of evidence and one kind of target, but each also leaves predictable blind spots. No single family can exhaust explanation quality on its own.

4 Comparative Synthesis and Open Gaps

4.1 Search Protocol and Corpus Profile

Search strategy. We searched five bibliographic databases: Semantic Scholar, Google Scholar, ACM Digital Library, IEEE Xplore, and ACL Anthology. Two query families were applied. The primary query combined evaluation-object terms (*"explainable AI" OR "XAI" OR "interpretable machine learning"*) with evaluation-act terms (*"evaluation" OR "benchmark" OR "metric" OR "assessment"*). A secondary query targeted method-comparative literature: (*"LIME" OR "SHAP" OR "attribution method"*) AND (*"evaluation" OR "stability" OR "fidelity"*). Both queries were restricted to English-language publications from January 2015 to March 2026. Conference proceedings, journal articles, and ArXiv preprints

Table 4: Corpus construction screening record. Identification and screening counts are from the original search log. The final two rows reflect the coded corpus released with this paper (`docs/reports/paper_bc/paper_bc_review_corpus.csv`), which was reconstructed from the citation record and the second-reviewer audit; see §7.

Stage	Records
Identified via database searches	312
Removed after cross-database deduplication	65
Screened on title and abstract	247
Excluded at screen (out of scope, non-English, no eval component)	152
Full text assessed for eligibility	95
Excluded at full text (intrinsic only, no codeable metric)	51
Included in coded corpus	**44**

with an accompanying peer-reviewed venue were eligible; informal workshop papers without self-contained empirical evaluations were excluded.

Inclusion and exclusion criteria. A paper was *included* if it: (i) proposes, operationalizes, or critically analyses an evaluation metric or evaluation paradigm for post-hoc model-agnostic explanations; (ii) provides a formal or empirical evaluation component beyond merely applying an explainer to a dataset; and (iii) is available in full text. A paper was *excluded* if it: focuses exclusively on intrinsic or white-box interpretability without a metric generalizable to post-hoc settings; is a domain-specific clinical or scientific application with no reusable evaluation component; or contributes only to model training methodology. Borderline cases were resolved by conservatively admitting sources containing at least one codeable evaluation construct relevant to the four-axis taxonomy.

Structured screening record. Table 4 documents the staged screening. Records were first pooled and deduplicated across databases, then screened on title and abstract, then assessed in full text. The record borrows the staged accounting logic of systematic reviews, but the review component is a structured scoping synthesis rather than a full systematic review.

Corpus structure. Each of the 44 coded papers is assigned to one or more thematic clusters and coded along the same four taxonomy axes introduced in Section 3 (evaluation target, evidence source, quality property, task context). Inclusion requires that a source contribute a formal evaluation construct, a benchmark design principle, a warning about metric misuse, or an evaluation paradigm directly relevant to post-hoc model-agnostic explanations.

Independent second-reviewer audit. The initial title/abstract screen and coding pass were conducted by the primary author. To stress-test the taxonomy before broader adjudication, an independent human second reviewer, recruited through the project's human-evaluation dashboard, audited a stratified subset of 16 papers (36% of the coded corpus),

Table 5: Descriptive profile of the review corpus (44 coded papers).

Profile item	**Value**
Corpus freeze	April 2026
Databases searched	5 (Semantic Scholar, Google Scholar, ACM DL, IEEE Xplore, ACL Anthology)
Records screened	247 (post-deduplication; see Table 4)
Unique coded papers	44
Cluster distribution	15 faithfulness/robustness, 10 taxonomy/survey, 7 human-grounded, 6 benchmark/toolkit, 4 LLM-judge, 2 counterfactual/recourse
Evidence-source coverage	21 proxy, 15 expert/taxonomy, 13 benchmark, 12 end-user, 4 LLM-judge (papers coded under multiple sources; 65 codes over 44 papers)
Confidence mix	44 high (full text held for every coded paper)

sampled across the six thematic clusters reported in Table 5. The audit records agreement on inclusion status and on the four taxonomy axes defined in Section 3; detailed results are reported in Supplementary Table S4. Because several axes are multi-label, agreement is summarized as exact agreement and Jaccard similarity, with Cohen's κ reserved for single-label screening decisions when both positive and negative cases are present. The audit produced 100% inclusion agreement on the audited included records, but lower exact agreement for the taxonomy axes, especially quality-property coding (25.0% exact agreement; mean Jaccard 0.657). We therefore treat the exercise as a bias-detection and codebook-refinement step on a subset of the corpus, not as a substitute for a full-corpus independent human inter-rater review.

The evidence-source distribution in Table 5 reveals a proxy skew: proxy evidence is coded for 21 of the 44 papers and is the primary source for 14 of them, the largest single category; benchmark-grounded evidence is coded for 13 papers and end-user evidence for 12. Structural metrics are therefore comparatively well standardized, while the meaning-oriented constructs remain thinly operationalized relative to the attention they receive.

4.2 Gap 1: Proxy Metrics Are Necessary but Not Sufficient

Faithfulness, stability, and sparsity remain essential because they create reproducible, quantitative anchors for benchmark comparison (Agarwal et al., 2022; Hedstrom et al., 2023; Canha et al., 2025; Nauta et al., 2023; Burkart and Huber, 2021). However, the coded corpus shows how dominant these proxies have become: proxy evidence is coded for 21 of the 44 studies and is the primary source for 14, more than any other category, while benchmark-grounded evidence is coded for 13. Recent work confirms that proxy-only evaluation can be misleading: attribution methods that score well on perturbation proxies may still fail sanity checks (Adebayo et al., 2018), be sensitive to hyperparameters (Alvarez-Melis and Jaakkola, 2018), or be deliberately manipulated (Slack et al., 2020). Proxy metrics do not directly answer whether an explanation reflects domain logic, causal relevance, or user-meaningful interpretation (Kumar et al., 2020; Doshi-Velez and Kim, 2017). The paired empirical study in Section 5 addresses precisely this proxy layer, filling the gap with controlled, auditable evidence for LIME and SHAP.

4.3 Gap 2: Human-Grounded Constructs Are Underspecified

The literature frequently invokes interpretability, clarity, plausibility, and usefulness, but these are often under-operationalized. When human studies exist, their tasks, rubrics, and participant populations vary so widely that direct comparison is difficult (Ali et al., 2023; Longo et al., 2024; Mohseni et al., 2021). In the coded corpus, 15 studies use expert-structured or taxonomy-level evidence and 12 use end-user evidence, but only 6 make end-user evidence their primary source. User studies further show that data scientists and ML practitioners engage with interpretability tools in ways that diverge from how tool designers intend them to be used (Kaur et al., 2020; Hase and Bansal, 2020; Buçinca et al., 2020). This weakens cumulative progress because the same label may refer to different constructs across papers—which is why Table 2 is necessary even for a survey of this scope. The framework proposed by Doshi-Velez and Kim (2017) remains the clearest call for rigorous, task-level evaluation, yet most subsequent benchmarking has reverted to proxy evidence.

4.4 Gap 3: Semantic Evaluation Is Rising Faster than Its Validation Base

LLM-based judging offers a practical way to score explanation quality at scale, especially when explanations or rubrics are textual. But the LLM-as-a-judge literature shows that proxy judges can display position bias, verbosity bias, or agreement illusions (Zheng et al., 2023; Gu et al., 2024). In XAI, semantic evaluator outputs should be treated as calibrated proxies rather than as self-validating replacements for expert review. The coded corpus contains 4 LLM-validation-relevant entries out of 44, one of which is itself a survey. That empirical base remains thin relative to the uptake of LLM evaluation in practice.

Negative result: LLM-judge inter-rater reliability (EXP4). To empirically ground this concern, we conducted a multi-rater reliability study (EXP4) in which three LLM judges independently scored $n = 147$ paired explanation outputs on seven semantic dimensions using structured rubrics (rubric template archived at `src/prompts/templates/explanation_eval.j2` in the companion repository). Table 6 reports intra-class correlation ICC(2,1) and Krippendorff's α for each dimension. ICC(2,1) requires a complete cases-by-judges matrix and is therefore computed on the $n = 147$ fully-triple-rated cases; Krippendorff's α tolerates missing ratings and is computed on the fuller $n = 192$-case pool, which includes cases with at least one missing judge score. All values fall below the conventionally accepted reliability threshold of ≥ 0.75 for *good* agreement (Koo and Li, 2016), confirming that semantic evaluation at this scale lacks the inter-rater convergence needed for confirmatory inference. We treat this as a negative result: LLM judges, under the rubrics provided, do not yet meet the psychometric bar required to serve as reliable proxies for expert semantic annotation in XAI. Completeness and semantic plausibility show the highest consistency ($\alpha > 0.55$), while clarity and audit usefulness are weakest ($\alpha < 0.35$), indicating the most semantic ambiguity in those constructs.

Table 6: LLM-judge reliability on seven semantic dimensions (EXP4; ICC on $n = 147$ fully-rated cases, Krippendorff's α on $n = 192$ cases (see text); 3 LLM judges). ICC(2,1) = two-way random-effects absolute-agreement model. No dimension reaches the ≥ 0.75 good-reliability threshold (Koo and Li, 2016).

Dimension	ICC(2,1)	Krippendorff α
Clarity	0.321	0.278
Concision	0.453	0.409
Actionability	0.400	0.362
Audit Usefulness	0.376	0.340
Completeness	0.585	0.566
Semantic Plausibility	0.601	0.573
Overall Quality	0.431	0.394

4.5 Formalization-Aware Evaluation Criteria

We extend the semantic evaluation scope from linguistic coherence to methodological adequacy by introducing formalization-aware criteria. Specifically, a rigorous evaluation should verify whether a studied method: (i) states a concrete stakeholder information need; (ii) specifies an explicit explanation problem; (iii) defines or approximates a correctness criterion; (iv) separates correctness from secondary desiderata such as robustness; and (v) provides evidence aligned with the stated criterion (Haufe et al., 2026; Sokol and Flach, 2020). This design aims to reduce false confidence induced by well-written but weakly specified explanation claims.

The motivation is that the XAI literature may contain claim–evidence mismatches, where broad interpretability conclusions are drawn from metrics that do not fully validate the intended explanatory use (Haufe et al., 2026; Hedstrom et al., 2023). Formalization-aware criteria assess claim discipline, assumption transparency, and evidence alignment. In this sense, they constitute a structured meta-evaluation layer intended to support reproducible synthesis rather than to replace domain expert judgment. Human review remains necessary for high-stakes interpretation (Buçinca et al., 2020; Bansal et al., 2021; Fok and Weld, 2023).

4.6 Layered Evaluation Architecture

The synthesis argues for layered evaluation. Proxy metrics should anchor reproducibility and comparative benchmarking. Human-subject or expert layers are required when the claim concerns user usefulness, task outcomes, plausibility, or causal adequacy. The present paper maps those layers and tests an LLM-proxy reliability condition, but it does not conduct a human-subject user study. The mistake is not using one family or another; it is allowing any single family to monopolize the definition of explanation quality.

Figure 1 summarizes the evaluation continuum. Benchmark evidence can support comparative technical claims. Semantic evaluator outputs can support exploratory interpretation until calibrated. Human agreement and task-outcome evidence are required before semantic or user-centered constructs can be treated as confirmatory claims.

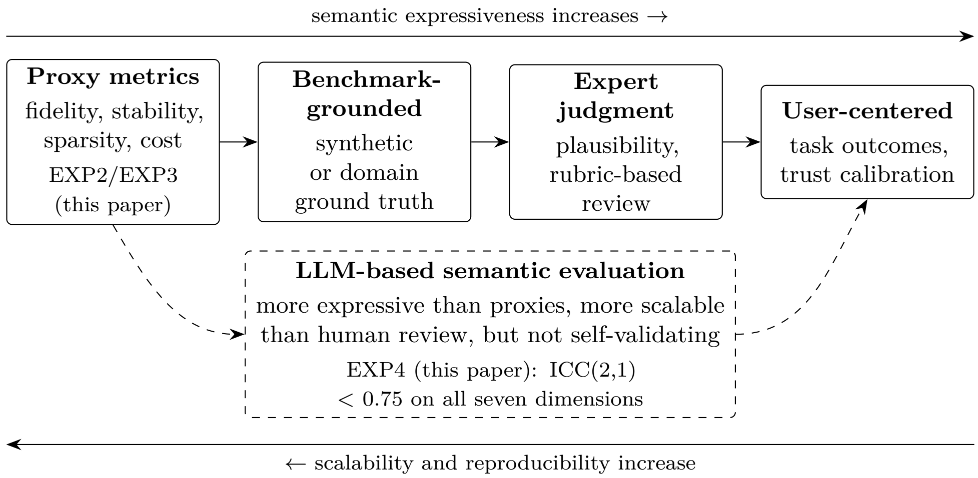

Figure 1: Evaluation continuum for post-hoc XAI, with representative metric families mapped to each evidence layer. LLM-based semantic evaluation occupies an intermediate position between proxy metrics and human review; the EXP4 reliability result (Section 4) shows it is not yet self-validating.

5 Empirical Benchmark: LIME versus SHAP

Section 4 established that proxy metrics dominate the current evidence base while human-grounded constructs remain underspecified. The present section fills the proxy layer with a controlled, paired comparison of LIME and SHAP—the two most widely deployed model-agnostic post-hoc explainers for tabular data. It does not test whether users make better decisions with either explainer. The study is designed to produce auditable, scope-bounded evidence on quality-cost trade-offs under controlled conditions, directly instantiating the "proxy and robustness" cell of the layered evaluation architecture.

5.1 Research Questions and Hypotheses

We address four directional questions, each paired with its testable directional hypothesis (Table 7).

Sparsity (P) is measured as active-feature ratio.

5.2 Experimental Design and Analysis Units

The benchmark is instantiated from a robustness cohort with factors:

- model family $g \in \{\text{logreg}, \text{rf}, \text{xgb}, \text{svm}, \text{mlp}\}$,
- explainer $k \in \{\text{LIME}, \text{SHAP}\}$,
- seed $s \in \{42, 123, 456, 789, 999\}$,
- sampling intensity $N \in \{50, 100, 200\}$ per TP/TN/FP/FN stratum.

Table 7: Research questions and corresponding directional hypotheses.

RQ	Question	Hypothesis
RQ1 (Robustness)	Is SHAP more stable than LIME under input perturbation?	$H_1 : S_{\text{SHAP}} > S_{\text{LIME}}$
RQ2 (Parsimony)	Is LIME sparser than SHAP?	$H_2 : P_{\text{LIME}} > P_{\text{SHAP}}$
RQ3 (Efficiency)	Is LIME computationally cheaper than SHAP?	$H_3 : C_{\text{LIME}} < C_{\text{SHAP}}$
RQ4 (Faithfulness)	Does SHAP provide stronger faithfulness and fidelity signals than LIME?	$H_4 : F_{\text{SHAP}} > F_{\text{LIME}}$

This yields $5 \times 2 \times 5 \times 3 = 150$ planned runs, all of which are artifact-qualified (75 LIME, 75 SHAP), forming 75 matched SHAP-LIME cells on identical (g, s, N) coordinates for paired inference.

To avoid pseudoreplication, we use a hierarchical analysis structure:

1. **Instance level:** per-case metric values.

2. **Run level:** mean metric per configuration.

3. **Matched-cell level:** paired SHAP-LIME run means on shared (g, s, N) cells (primary inferential unit).

5.3 Data, Preprocessing, and Model Controls

All experiments use the UCI Adult dataset (Kohavi, 1996) with a stratified split and deterministic seeds. The preprocessing pipeline follows train-only fitting with numeric scaling (`StandardScaler`) and categorical one-hot encoding (`OneHotEncoder`; unknowns ignored). A fixed set of pre-trained black-box models is shared across all explainer runs to isolate explainer effects: random forest (Breiman, 2001) and gradient-boosted trees (Chen and Guestrin, 2016), alongside logistic regression, SVM, and MLP families. Evaluation instances are sampled by error quadrants (TP, TN, FP, FN) via stratified sampling. Realized instance counts range from 27 to 800 (median 400).

5.4 Explainer Protocol

Both explainers consume the same transformed feature space and target the positive class probability:

- **SHAP:** TreeExplainer for tree models, KernelExplainer for non-tree models, with background subsampling ($n = 50$).

- **LIME:** tabular local surrogate with `num_features=10`, effective `num_samples=1000`, and `kernel_width=3.0`.

5.5 Metric Operationalization

Primary metrics are computed per instance and aggregated per run:

1. **Stability** S: mean pairwise cosine similarity of attribution vectors under Gaussian perturbations ($T = 15$, $\sigma = 0.1$):

$$S = \frac{2}{T(T-1)} \sum_{a<b} \cos\left(e^{(a)}, e^{(b)}\right).$$

2. **Sparsity** P: active-feature ratio above threshold $\tau = 10^{-4}$ (lower active ratio indicates sparser explanations).

3. **Fidelity** F: faithfulness correlation between attribution magnitude and prediction drop under feature masking.

4. **Faithfulness Gap** Δ_k: absolute prediction change after masking the top-k features ($k = 5$):

$$\Delta_k = \left| f(x) - f(x_{\text{mask-top-k}}) \right|.$$

5. **Cost** C: per-instance wall-clock explanation time (ms).

Sparsity operationalization note. LIME's active-feature count is governed by the `num_features=10` hyperparameter, which acts as a hard ceiling on the number of non-zero attributions returned. This means LIME sparsity is a *hyperparameter constraint*, not an emergent property of the explanation model. The active-feature ratio metric therefore measures whether LIME, *as configured*, returns fewer significant features than SHAP; it does not test whether LIME is intrinsically sparser under unconstrained settings. Readers comparing parsimony results across studies should verify that equivalent `num_features` bounds are applied, since the direction and magnitude of the SHAP–LIME parsimony gap will change with different settings.

5.6 Inferential Protocol

The statistical plan follows established comparative-ML practice (Demsăr, 2006):

1. primary test: two-sided paired Wilcoxon signed-rank on matched SHAP-LIME cells for each metric (Wilcoxon, 1945),

2. effect size: paired d_z for magnitude reporting (Lakens, 2013),

3. multiplicity control: Holm-Bonferroni over the five primary metrics (Holm, 1979).

All inferential claims are restricted to artifact-qualified runs: parseable, non-empty results with explicit matched pairing. No semantic endpoint enters the confirmatory analysis reported here.

Table 8: Friedman omnibus test across four methods (EXP2, $n = 15$ blocks — 5 model families × 3 sampling intensities, each a 5-seed mean — $k = 4$). Reported p-values are Holm-Bonferroni-adjusted across the five primary metrics; the corresponding raw values are 3.78×10^{-9} (fidelity) and 7.65×10^{-9} (stability). W = Kendall's concordance coefficient.

Metric	χ^2 (df = 3)	p_{Holm}	W
Fidelity	42.12	1.51×10^{-8}	0.936
Stability	40.68	2.29×10^{-8}	0.904

5.7 Reproducibility Contract

The protocol is fully configuration-driven (YAML with seeded execution), with per-run `results.json` artifacts, explicit exclusion of malformed or empty runs, and no synthetic imputation. This provides a deterministic audit trail from configuration to manuscript-level claim.

5.8 Results

Results are reported from the artifact-qualified robustness cohort. The analysis covers 150 analyzable runs (75 LIME, 75 SHAP) forming 75 paired cells with identical (model, seed, N). Matched coverage spans all five model families (xgb = 15, logreg = 15, rf = 15, mlp = 15, svm = 15) and all sampling intensities with near-balanced counts.

Multi-Method Context (Friedman)

As pre-analysis context, we ran a Friedman test across all four methods in EXP2 (SHAP, LIME, Anchors (Ribeiro et al., 2018), DiCE (Mothilal et al., 2020)) on $n = 15$ blocks per method, one per (g, N) model-family-and-sampling-intensity combination, each block value the mean across the 5 available seeds. Block-level aggregation was necessary because Anchors and DiCE runs are not complete across the full (g, s, N) grid (57/75 and 68/75 artifact-qualified runs, respectively, versus 75/75 for SHAP and LIME). The full EXP2 cohort covers all four methods; the paired inference below is restricted to the LIME-SHAP matched subset for confirmatory claims. Table 8 shows highly significant systematic differences in both fidelity and stability (Kendall's $W > 0.90$ in both cases), establishing that the four-method field is not exchangeable.

Method mean values and average ranks are shown in Table 9. Nemenyi post-hoc testing (CD = 1.211, $\alpha = 0.05$) confirms that SHAP significantly outperforms Anchors and DiCE in fidelity (rank gaps 2.2 and 2.8), and significantly outperforms LIME and Anchors in stability (rank gaps 2.4 and 2.6). The SHAP–DiCE stability rank gap (1.0) falls below the CD threshold (Nemenyi $p = 0.15$), consistent with the comparative run-to-run consistency of DiCE's counterfactual outputs; SHAP's stability advantage over DiCE is therefore descriptive rather than confirmed at the omnibus level. The SHAP-LIME omnibus fidelity rank gap equals exactly 1.0—marginally below the CD threshold—but the directional paired

Table 9: Mean metric values by method from EXP2 (mean of 15 model×sampling-intensity block means; underlying artifact-qualified run coverage: 75/75 SHAP and LIME, 68/75 DiCE, 57/75 Anchors) with Nemenyi average ranks. CD = 1.211; pairs with rank gap > CD are significant at $\alpha = 0.05$.

	SHAP	LIME	Anchors	DiCE
Fidelity (mean)	0.808	0.560	0.389	0.170
Stability (mean)	0.732	0.014	0.043	0.361
Fidelity rank	1.0	2.0	3.2	3.8
Stability rank	1.0	3.4	3.6	2.0

Table 10: Matched SHAP-versus-LIME paired results ($n = 75$ cells). For all metrics, $\Delta =$ SHAP − LIME. Sparsity is measured as active-feature ratio; larger values indicate denser explanations.

Metric	LIME Mean	SHAP Mean	Δ	p_{Holm}	d_z
Stability	0.014	0.732	+0.718	2.6×10^{-13}	3.00
Sparsity (active ratio)	0.085	0.226	+0.142	2.6×10^{-13}	0.84
Fidelity	0.560	0.808	+0.248	2.6×10^{-13}	4.82
Faithfulness Gap	0.334	0.380	+0.045	2.6×10^{-13}	2.63
Cost (ms)	3660.7	11708.3	+8047.6	1.2×10^{-10}	0.45

Wilcoxon below demonstrates significance when power is concentrated on this specific two-method comparison.

Primary Paired Inference

Table 10 summarizes paired run-level outcomes on the 75 matched cells. All five primary endpoints remain significant under Holm-adjusted testing.

The stability effect is the largest practical separation in the study. Median paired stability difference is +0.866 in favor of SHAP, and the direction is consistent in all 75 matched cells. This indicates that LIME's local surrogate sampling introduces substantial run-to-run volatility under perturbation, whereas SHAP is comparatively invariant.

For sparsity, the sign is also consistent in all 75 matched cells, but interpretation requires care: the metric here is active-feature ratio. SHAP's larger value means denser explanations; LIME is therefore systematically sparser and easier to compress for user-facing presentation.

Fidelity favors SHAP in all 75 matched cells, with a mean paired difference of +0.248 and a very large effect size ($d_z = 4.82$). Faithfulness gap also favors SHAP in 74 of 75 matched cells, with a mean paired difference of +0.045 and a large effect size ($d_z = 2.63$). These results indicate not only statistical separation but practically meaningful quality

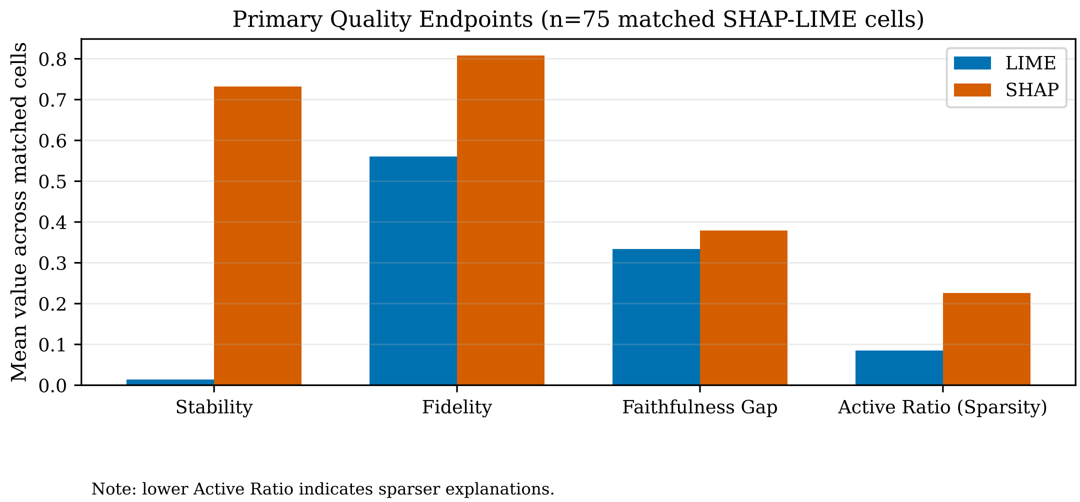

Figure 2: Paired mean comparison across quality-oriented primary endpoints ($n = 75$ matched SHAP-LIME cells). Active ratio is reported directly; lower values imply sparser explanations.

Table 11: Median coefficient of variation (CV = SD/mean) across seeds per (N, model) stratum for the two primary quality endpoints.

Method – Metric	Median CV	Reproducible?
SHAP – Fidelity	0.8%	Yes
SHAP – Stability	0.7%	Yes
LIME – Fidelity	2.6%	Yes
LIME – Stability	86.2%	No

gains under this protocol. Figure 2 visualizes the paired mean separations for the four quality-oriented endpoints.

Reproducibility Probe

To characterize measurement stability under seed variation, we computed the coefficient of variation (CV = SD/mean) across the five seeds for each (N, model) stratum and report the stratum-median CV in Table 11. SHAP yields highly reproducible fidelity and stability readings (CV $< 1\%$). LIME fidelity is acceptable (CV $= 2.6\%$), whereas LIME stability is structurally irreproducible (CV $= 86.2\%$). This is not a measurement artifact: when the mean cosine similarity is ≈ 0.014, any nonzero variation produces an extreme relative deviation. The CV probe confirms that LIME stability is geometrically unstable—not merely noisy.

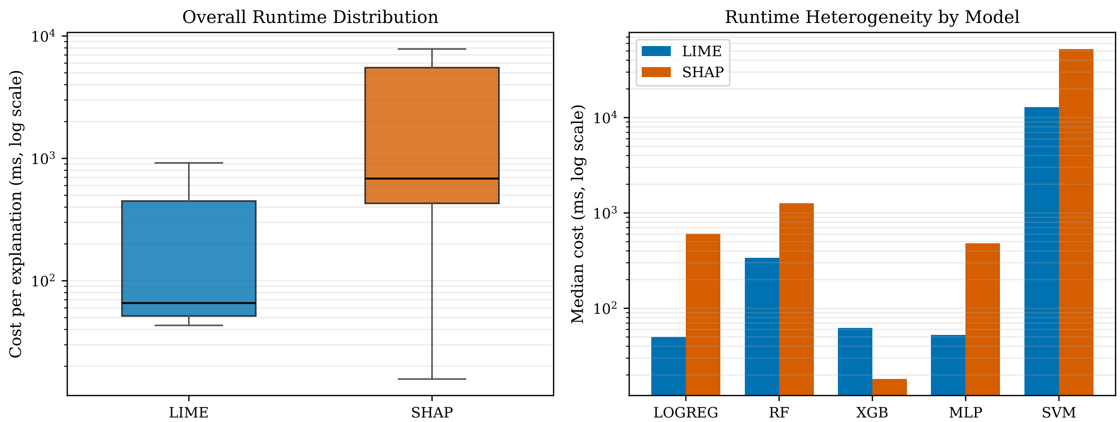

Figure 3: Runtime behavior under matched-cell pairing ($n = 75$). Left: overall cost distribution by method (log scale). Right: model-level median cost (log scale), showing the XGBoost exception where SHAP is faster while other families are SHAP-slower.

Runtime Distribution and Heterogeneity

The cost signal is significant but strongly heterogeneous. SHAP is slower in 59 of 75 matched cells. Matched-cell median cost is 684.1 ms for SHAP versus 65.7 ms for LIME, corresponding to a median SHAP/LIME cost ratio of approximately 5.3×. Means are more separated (11,708.3 ms vs. 3,660.7 ms) because the SHAP distribution is heavy-tailed: the 90th and 95th percentile latencies are approximately 51,456 ms and 67,066 ms, respectively, compared with 12,825 ms and 13,192 ms for LIME. This tail behavior is concentrated in kernel-SHAP contexts, especially SVM. Figure 3 shows both the overall log-scale runtime distribution and model-level median heterogeneity.

Heterogeneity analysis by model family shows that quality ordering (stability, sparsity-direction, fidelity, faithfulness gap) is robust, while cost ranking is model-dependent. SHAP is faster in all 15 XGBoost matched cells and in one SVM cell, whereas LogReg, RF, and MLP are uniformly SHAP-slower. Therefore, “LIME is faster” is a majority statement in this cohort but is not universally true across model classes.

Synthesis of Trade-offs

The empirical frontier is clear: SHAP delivers higher reliability and faithfulness at substantially higher expected runtime, while LIME delivers lower latency and stronger compression at the cost of robustness. The results therefore support objective-driven selection rather than global method supremacy. For assurance-critical analysis, SHAP’s quality margin is large and consistent. For interaction-critical systems, LIME’s runtime profile remains operationally attractive.

Table 12: EXP3 external validation: mean fidelity (3-seed average) for SHAP vs. Anchors across two additional datasets. SHAP > Anchors in all 12 paired combinations. SHAP values are taken from the current committed `experiments/exp3_cross_dataset/results/` snapshot, which is also the cohort Paper A reports.

Dataset	Model	SHAP	Anchors
Breast Cancer	RF	0.779	0.265
Breast Cancer	XGB	0.617	0.208
German Credit	RF	0.714	0.351
German Credit	XGB	0.711	0.451

5.9 External Validation: Cross-Dataset Replication (EXP3)

To partially probe tabular external validity, we ran a cross-dataset validation study (EXP3) comparing SHAP, LIME, and Anchors (Ribeiro et al., 2018) on two additional binary classification datasets: Breast Cancer Wisconsin (Dua and Graff, 2019) ($n = 569$, 30 continuous features) and German Credit (Dua and Graff, 2019) ($n = 1{,}000$, 20 mixed features). Both used random forest and XGBoost models across three random seeds each, yielding 12 paired dataset–model–seed comparisons per method pair.

Fidelity ordering (SHAP vs. Anchors). SHAP outperforms Anchors in fidelity in all 12 paired combinations (Table 12, Figure 4), with mean gaps ranging from +0.26 (German Credit, XGB) to +0.51 (Breast Cancer, RF). This provides partial tabular support for the EXP2 fidelity direction across different feature spaces, dimensionalities, and class distributions.

LIME cross-dataset extension. Table 13 reports LIME fidelity and stability on the same datasets. Two findings are noteworthy. First, SHAP > LIME fidelity holds consistently on German Credit (gaps +0.23–+0.24 for RF, +0.24–+0.25 for XGB), replicating the EXP2 direction. On Breast Cancer the fidelity gap narrows substantially: SHAP leads LIME by only +0.07 (RF) and +0.07 (XGB), compared to +0.27 (RF) and +0.21 (XGB) in the corresponding model strata on Adult. Second, and more critically, LIME *stability* on these datasets (0.85–0.93) is an order of magnitude higher than on Adult (0.014). This reveals that the near-zero stability observed in EXP2 is not a fixed structural property of LIME: it is strongly modulated by feature-space characteristics. The Adult dataset's 103-dimensional one-hot-encoded space appears to be the primary driver of instability; in lower-dimensional spaces with more continuous features, LIME produces substantially more consistent explanations.

Scope limitation. SHAP stability on the cross-dataset cohorts was not instrumented in EXP3; a direct SHAP–LIME stability comparison on Breast Cancer and German Credit remains pending. The finding that LIME stability varies dramatically with feature-space dimensionality and encoding density should be taken as a moderation hypothesis requiring further confirmation across a broader set of datasets and preprocessing schemes.

Table 13: EXP3 LIME extension: mean fidelity and stability (3-seed average) for LIME on the two cross-dataset cohorts. Stability = mean pairwise cosine similarity under $T = 15$ Gaussian perturbations ($\sigma = 0.1$); same protocol as EXP2. For comparison, LIME on Adult (EXP2) in the matched model strata: fidelity 0.457 (RF) and 0.553 (XGB); stability $\approx$ 0.01–0.02 across all model families.

Dataset	Model	Fidelity	Stability	Sparsity	Cost (ms)
Breast Cancer	RF	0.713	0.922	0.331	12.3
Breast Cancer	XGB	0.543	0.927	0.331	7.5
German Credit	RF	0.483	0.854	0.163	21.1
German Credit	XGB	0.468	0.748	0.163	10.2

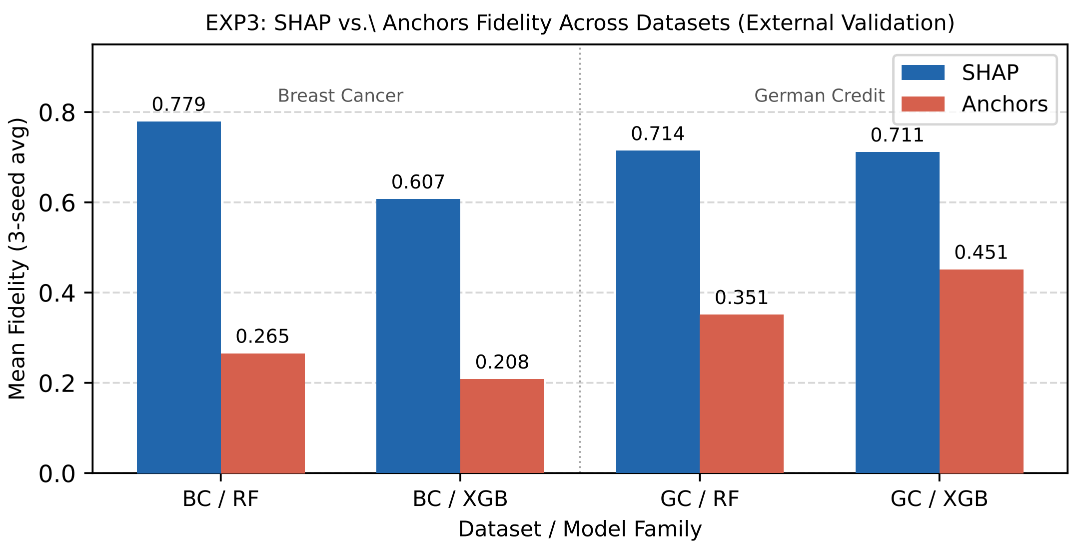

Figure 4: EXP3 external validation: mean fidelity (3-seed average) for SHAP vs. Anchors across four dataset-model combinations. SHAP exceeds Anchors in all 12 paired (dataset $\times$ model $\times$ seed) combinations. BC = Breast Cancer; GC = German Credit.

6 Practical Recommendations

Based on the taxonomy (Section 3), the gap analysis (Section 4), and the empirical findings (Section 5), we propose the **Hybrid Deployment Pattern**:

This two-tier pattern directly reflects the layered evaluation architecture identified in Section 4: proxy and benchmark evidence (Tier 2) can be complemented by interaction-optimised lightweight explanations (Tier 1), each operating within its validated scope. Critically, the Tier 1 criterion is not a blanket method preference but a function of model architecture: the existence of a model-aware explainer (e.g., `TreeExplainer`) collapses the

Table 14: Hybrid Deployment Pattern: tier-conditioned explainer recommendation.

Tier	Recommendation	Evidence and recommended condition
Tier 1: Real-Time / End-User	Model-architecture-conditioned: **SHAP** (`TreeExplainer`) for tree-based models (XGBoost, LightGBM, Random Forest); **LIME** for non-tree/black-box models (neural networks, opaque APIs).	For tree models, exact Shapley values in $O(TLD)$ time undercut LIME's sampling loop, so SHAP is simultaneously faster *and* more faithful and the usual fidelity–latency trade-off does not apply. For non-tree models, LIME's lower median runtime (53.3 ms vs. 694.6 ms for KernelSHAP; Figure 3) gives a latency advantage where sampling is the only option. Recommended when sub-second latency and compact explanations matter more than maximizing fidelity and no model-aware explainer is available.
Tier 2: Auditing / Data Science	**SHAP.**	Use for bias analysis, model debugging, or regulatory inquiries: consistent paired advantages in stability, fidelity, and faithfulness gap (d_z = 3.00, 4.82, 2.63). Recommended for analyst-mediated or offline workflows tolerating second-to-minute latency; plan sample budgets according to whether SHAP runs in tree-specific or kernel mode.

fidelity–latency trade-off, making SHAP the dominant choice at both tiers for tree-based deployments. This maps directly onto the Task Context axis of the taxonomy (Section 3): explainer selection is determined first by architectural affordance, then by deployment constraint.

7 Validity Boundaries and Limitations

7.1 Interpretation Scope of SHAP–LIME Comparisons

The SHAP–LIME differences reported here should be interpreted as comparative performance within our benchmark protocol, not as definitive evidence that one method is universally more correct. Recent formal analyses show that multiple attribution families can produce high-importance assignments to suppressor variables in dependent-feature settings, complicating direct interpretation of attribution magnitude as target-relevant importance (Wilming et al., 2022, 2023; Haufe et al., 2026). Observed performance gaps between SHAP and LIME may therefore reflect both methodological differences and sensitivity to data dependence structure.

We distinguish *ranking claims* from *validity claims*. A ranking claim states that one explainer outperforms another on predefined benchmark criteria in this dataset and protocol. A validity claim would require stronger evidence that explanations satisfy a formal correctness criterion for the intended task, potentially including controlled synthetic or transparent-model ground-truth tests, dependency-aware perturbation protocols, human-centered validation, and explicit assumption checks (Haufe et al., 2026; Doshi-Velez and Kim, 2017). This paper primarily makes ranking claims; broader validity claims remain provisional.

This framing aligns with prior findings that faithfulness-style evaluations can be informative but are not sufficient to resolve correctness questions in isolation (Jacovi and Goldberg, 2020; Adebayo et al., 2018; Hooker et al., 2019; Rong et al., 2022). The SHAP–LIME comparison presented here is best read as robust comparative evidence under the evaluated conditions, with external validity depending on how closely downstream applications match those conditions.

7.2 Taxonomy Scope and Single-Reviewer Limitation

Corpus provenance. The coded corpus released with this paper was reconstructed. The original coding pass was not preserved in the project repository, and the corpus was rebuilt from two surviving sources: the second-reviewer audit record, which carries the original first-reviewer coding on all four axes for 16 papers, and the paper's own citation record, from which a further 28 coded papers were identified and re-coded from their full text. Approximately four papers that were coded but never cited could not be recovered, so the released corpus contains 44 rather than the 48 originally coded, and the distributions in Table 5 are those of the reconstruction rather than of the original pass. The qualitative gap analysis is unchanged by this, but readers should treat the corpus counts as reconstructed figures with a documented provenance rather than as a preserved primary record.

The survey employs a structured database search with documented inclusion and exclusion criteria and a transparent screening record (Table 4). It is not, however, a full systematic review: the search used five databases rather than an exhaustive database set, and the initial title/abstract screen and coding pass were performed by a single reviewer. The independent second-reviewer audit described in Section 4 covers only a stratified subset of the corpus and identifies likely coding ambiguities, but it does not replace a full-corpus independent human inter-rater review pipeline. The emphasis on model-agnostic post-hoc XAI reflects the scope of the empirical study; intrinsic interpretability and some modality-specific traditions are covered only insofar as they affect metric transfer. The taxonomy organizes construct space—not effect sizes—and does not estimate a pooled numerical ranking across studies.

7.3 Statistical and Construct Validity

Confirmatory claims are restricted to the 75 fully matched SHAP-LIME cells in the robustness cohort. All 75 LIME and 75 SHAP runs are artifact-qualified and available for paired inference. Reported metrics are implementation-bound rather than abstract labels; their operational definitions are anchored in the companion repository source files for fidelity, stability, sparsity, faithfulness, and cost. No semantic, human-utility, or LLM-judge claim enters the confirmatory scope.

Axiomatic properties and metric alignment. SHAP satisfies four axioms—dummy, linearity, symmetry, and efficiency—that guarantee attributions distribute the full prediction change $f(x) - \mathbb{E}[f]$ as uniquely fair marginal contributions. Perturbation-based fidelity metrics measure whether high-magnitude attributions correspond to large prediction drops under feature masking. These two perspectives are structurally compatible: both quantify how faithfully attribution magnitudes encode feature-level causal responsibility for the model output. SHAP's axiomatic guarantee implies that, conditional on its background distribution assumptions, its attribution rankings are optimally aligned with the feature-removal signal that fidelity metrics probe. This does not mean SHAP axioms certify correctness in all deployment contexts—the axioms hold for the model, not for any latent data-generative mechanism—but it does explain why SHAP consistently outperforms LIME on perturbation-based fidelity without requiring that LIME violates a formal criterion.

7.4 Internal and External Validity

Runtime comparisons are protocol-specific. SHAP combines TreeExplainer for tree models and KernelExplainer for non-tree models, whereas LIME is run with fixed local-surrogate settings (`num_features=10`, effective `num_samples=1000`, `kernel_width=3.0`). Accordingly, the reported cost gap reflects both method family and concrete implementation pathway. External generalization is only partially probed by EXP3 (Section 5.9), which supports the tabular fidelity direction across Breast Cancer and German Credit but does not replicate the full SHAP–LIME endpoint set. SHAP–LIME comparisons and stability rankings remain bounded to the Adult cohort and may shift under different modalities, feature transformations, masking schemes, or hyperparameter settings.

LIME stability is not a convergence artifact. Two robustness probes rule out under-sampling and kernel miscalibration as explanations for the near-zero LIME stability (cosine similarity mean ≈ 0.014, CV = 86.2%). Varying `num_samples` $\in \{500, 1000, 2000\}$ under otherwise identical conditions (RF model, seed 42, $N = 100$, `kernel_width=3.0`) leaves stability near-zero and fidelity essentially flat at every level (Supplementary Table S5): doubling or halving the sample budget neither stabilizes the explanations nor changes the fidelity ordering. A companion `kernel_width` probe (Supplementary Table S2) shows that widths below 3.0 instead produce degenerate zero-weight explanations, confirming that the reference setting is the minimum viable configuration rather than a conservative choice. Both probes indicate the instability is structural to LIME's local surrogate fitting under Gaussian perturbation, not an artifact of insufficient compute.

Feature correlation and out-of-distribution masking. The Adult Income dataset contains moderate-to-strong inter-feature correlations (Supplementary Table S3) that affect the validity of perturbation-based fidelity metrics. When correlated features are independently zeroed or replaced by reference values during masking—as in our fidelity and faithfulness-gap computations—the resulting masked instances may fall outside the training distribution. Under such out-of-distribution (OOD) inputs, tree-based model outputs can be unreliable extrapolations, and the correlation structure between masked and retained features is ignored. This can cause fidelity metrics to either overestimate (if the model is

Table 15: Generalizability boundary of empirical claims.

Claim	Current evidence	Generalization status
SHAP > LIME stability	Adult Income matched cells only	Confirmed only for the Adult tabular protocol; EXP3 does not instrument SHAP stability
SHAP > LIME fidelity	Adult Income plus partial EXP3 tabular extension	Partially supported within tabular classification; not established for image, text, graph, or time-series data
LIME lower runtime	Adult Income, model-dependent	Bounded by model architecture and SHAP implementation; tree-specific SHAP reverses the ranking
SHAP > Anchors fidelity	EXP3 Breast Cancer and German Credit	Partial tabular replication of fidelity ordering, not a SHAP–LIME stability or runtime replication
Human-subject utility	No participant study; EXP4 is an LLM reliability probe only	Not established; user/task outcomes remain outside confirmatory scope
Non-tabular behavior	No image, text, graph, or time-series experiments	Out of scope for the present confirmatory benchmark

insensitive to OOD inputs) or underestimate (if the model reacts to implausible feature combinations) the true attribution quality.

To assess whether this concern changes the method ordering, we added EXP6, a top-feature masking-sensitivity probe over the Adult RF stratum. EXP6 reuses stored EXP2 top-10 explanations for the $n = 50$ RF runs across five seeds and recomputes prediction-drop metrics under four masking schemes: independent zero masking, mean replacement, marginal empirical replacement, and grouped mean replacement, where the grouped scheme masks correlated processed feature sets together (e.g., `education`/`education-num`). SHAP's top-k prediction-drop advantage remains positive in all five paired RF seed cells under every scheme, though its magnitude decreases under progressively more correlation-aware masking (full breakdown in Supplementary Table S6). Because EXP6 uses stored top-10 explanations rather than full attribution vectors, it should be read as a construct-validity stress test rather than a replacement for the primary full-run fidelity analysis: the RF fidelity direction is robust in this probe, while the effect magnitude is partly masking-protocol dependent.

7.5 Reproducibility and Operational Validity

All reported claims are traceable to seeded YAML configurations, per-run `results.json` artifacts, explicit exclusion of malformed or empty runs, and inferential exports in the companion repository. Figures are regenerated directly from the paired-cell Wilcoxon exports; no synthetic imputation is used at any stage. Before journal submission, the exact review snapshot should be frozen as a versioned release and archived under a version-specific DOI.

7.6 Future Work

Several limitations define natural extensions:

- **Dataset and modality scope:** The confirmatory SHAP–LIME comparison is bounded to tabular classification, with the strongest evidence coming from the Adult Income benchmark. EXP3 partially extends fidelity observations to Breast Cancer and German Credit, but it does not replicate the full SHAP–LIME endpoint set and does not test image, text, graph, or time-series explanations. Those modalities require different masking schemes, perturbation semantics, and evaluation endpoints.

- **Configuration space:** Both methods were run with default hyperparameter settings. The completed LIME `kernel_width` probe (Section 7, Supplementary Table S2) confirms the reference setting is a minimum-viable rather than conservative choice; SHAP background-set sensitivity and cross-model kernel-width effects remain open.

- **Runtime heterogeneity:** The XGBoost exception shows that "LIME is faster" is not a universal rule and warrants tree-specific analysis.

- **No human-subject validation:** This paper does not establish human-grounded usefulness, user comprehension, trust calibration, or decision-quality improvements. Validated user studies with task-outcome measures remain future work.

- **LLM-judge calibration:** EXP4 shows that ICC remains below 0.75 for all seven semantic dimensions; developing rubrics that achieve calibration-grade inter-rater agreement is an open challenge.

- **Taxonomy completeness:** The survey corpus is a working corpus rather than an exhaustive bibliometric sample; expanding it with systematic search protocols would strengthen the gap analysis.

- **Causal and counterfactual endpoints:** Integration of causal attribution methods and diverse counterfactual generation (Mothilal et al., 2020) remains an open follow-up.

8 Code and Artifact Availability

The manuscript, experimental code, and supporting materials are available in the public repository at `https://github.com/jonnabio/xai-eval-framework`, released under the MIT License. Artifacts needed to reproduce the empirical study include:

- EXP2 results: `experiments/exp2_scaled/results/`

- EXP2 inferential exports: `outputs/analysis/paper_a_exp2_stats/` (includes `analysis_summary.json`, `wilcoxon_shap_lime_all_models.csv`, `paired_cells_shap_lime_all_models.csv`)

- EXP3 results: `experiments/exp3_cross_dataset/results/` (with LIME extension exports in `outputs/analysis/exp3_lime_results.csv`)
- EXP4 ICC exports: `outputs/analysis/exp4_llm_evaluation/`
- EXP6 masking-sensitivity exports: `outputs/analysis/exp6_masking_sensitivity/`
- analysis scripts: `scripts/run_exp2_statistical_analysis.py`, `scripts/generate_paper_b_figures.py`, `scripts/run_exp6_masking_sensitivity.py`

The review corpus underlying the taxonomy and gap analysis is released as `docs/reports/paper_bc/paper_bc_review_corpus.csv`, with the retrieved full texts under `docs/reports/paper_bc/corpus_pdfs/` and per-paper provenance in `docs/reports/paper_bc/corpus_pdfs/RETRIEVAL_LOG.md`. Before journal submission, the exact snapshot will be frozen as a versioned release with a DOI.

9 Conclusion

XAI evaluation is not one problem but a family of related measurement problems. The literature has developed useful metric families for fidelity, robustness, sparsity, and benchmark reproducibility, yet these families do not fully capture semantic adequacy or human-centered meaning. By organizing model-agnostic post-hoc evaluation along the axes of evaluation target, evidence source, quality property, and task context, this paper clarifies why metric disagreement is common and why layered evaluation is necessary.

The main practical implication is twofold. First, benchmark metrics and semantic evaluation should be treated as complementary rather than competing forms of evidence. The taxonomy makes explicit where each metric family gains strength and where it leaves blind spots. Second, the choice between LIME and SHAP is not a matter of one being universally better but of fit for purpose. Under the paired benchmark protocol, SHAP is the preferred choice when stability and fidelity-oriented quality dominate the objective, while LIME remains the pragmatic option when latency and compactness are the primary constraints.

Our paired analysis across 75 matched Adult Income robustness cells shows consistent SHAP advantages in stability ($d_z = 3.00$), fidelity ($d_z = 4.82$), and faithfulness gap ($d_z = 2.63$), alongside systematic LIME advantages in sparsity and lower runtime for most model contexts. Cross-dataset replication (EXP3) provides partial tabular support for the fidelity direction on two additional datasets, while leaving stability, runtime, and non-tabular generalization unresolved. These results, read through the lens of the taxonomy, support a layered deployment architecture for tabular classification that separates user-facing low-latency explanation paths from high-assurance auditing workflows. This deployment architecture is derived from proxy metrics and runtime constraints; it should not be read as evidence that either method improves human task performance without a separate user study.

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